Sahaj Bhadja

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Summary

Full-Stack Software Engineer with 2+ years shipping **React**, **Node.js**, and **TypeScript** products, **LLM-powered** applications, and automated **CI/CD** deployments backed by PostgreSQL, MongoDB, and cloud infrastructure.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Web Technologies, Database Systems, Analysis of Algorithms, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *LA, California*

- Restructured a fault-tolerant async **LLM orchestration** system with FastAPI background workers, explicit job-state transitions, and failure isolation, processing 1K+ AI jobs and tripling throughput without cascading failures.
- Architected a multi-tenant AI education platform on Azure SQL and Blob Storage with **row-level security** for 500+ users, tuning indexed queries to cut dashboard latency from 1.2s to 350ms.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *LA, California*

- Built a full-stack AI grading platform with a **React** dashboard and modular Flask APIs, integrating GPT-4o, Claude, and Qwen-VL behind switchable providers to grade 10K+ daily text and handwritten submissions.
- Automated deployments to Render and Vercel through **GitHub Actions** CI/CD and added OCR fallbacks plus per-call token cost tracking, serving 150+ instructors while cutting feedback turnaround 60%.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) *India*

- **Automated** nightly Jenkins pipelines that built executables, provisioned VMs, and ran unit and integration tests across Windows, Linux, and macOS, reducing manual effort for cross-platform data-sharing microservices.
- Optimized large-scale APIs with pagination, field filtering, indexing, and caching for low-bandwidth users, improving response times 30% and leading an AWS Glue ETL redesign that boosted pipeline efficiency 68%.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) *India*

- Built a **MERN** resource platform with React, **Node.js**, Express, and MongoDB, using **WebSockets** for real-time allocation updates and synchronizing 2M+ records across locations through eight modular microservices.
- Automated Jenkins CI/CD with Docker packaging, staged rollouts, and zero-downtime production updates on Azure App Services, achieving 95%+ successful build validation while Sentry and Kibana surfaced errors.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Built Flask booking, pricing, and payment APIs behind React Native and **TypeScript** clients, using Firebase real-time sync to speed booking confirmations 40% while holding sub-200ms latency across 50K+ records.
- Automated payment webhooks for refunds, disputes, and status updates with idempotent handling and Firebase transactions preventing double bookings, reducing manual refund handling by about 70% across mobile and web.

Skills

Languages & Databases: TypeScript, JavaScript, Python, Java, C++, SQL, Bash, PostgreSQL, MongoDB, MySQL, Firebase
Full-Stack: React.js, Next.js, Node.js, Express.js, HTML, CSS, TailwindCSS, REST APIs, WebSockets, FastAPI, Flask
Cloud & Automation: GCP, AWS, Microsoft Azure, Docker, Kubernetes, CI/CD, GitHub Actions, Jenkins, Vercel, Linux, Git
AI & Systems: LLM Systems, OpenAI API, Prompt Engineering, FAISS, Transformers, PyTorch, Distributed Systems

Projects

Big-Brain Live | *LLaMA-2, FAISS, Pinecone, Convex, Clerk, GCP*

- Built an AI document platform with LLaMA-2 and FAISS semantic search, Clerk authentication, and Convex **real-time collaboration** on GCP, reaching 95% query accuracy across 50K+ documents for 200+ users.

Chat2SQL Code | *TypeScript, Java, Node.js, Docker, Maven, PostgreSQL*

- Extended an AI-powered **developer workspace** in TypeScript and Java that generates schema-aware SQL across 10+ databases, with Docker-based deployment and cross-platform desktop support on Windows, macOS, and Linux.

FinNexus Live | *Next.js, TypeScript, Kubernetes, MongoDB, WebSockets, Plaid*

- Built a Next.js and TypeScript fintech app with Plaid bank linking, WebSocket transaction updates, and **Kubernetes** deployments automated through CI/CD pipelines, supporting 50K+ users and 1M+ transactions.